RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105

DEPARTMENT OF INFORMATION TECHNOLOGY

CS23621 MOBILE APPLICATION DEVELOPMENT LABORATORY
Laboratory Record Note Book

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Academic Year :	……...2025-2026…………….

RAJALAKSHMI ENGINEERING COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: ....SRIMAN VIYASEN S J..

Academic Year: …2025-2026… Semester: …VI…… Branch: ……IT………

Register No. | **2116231001208**

Certified that this is the Bonafide record of work done by the above student in the CS23621- Mobile Application Development Laboratory during the academic year 2025 – 2026.

Signature of Faculty in-charge

Submitted for the Practical Examination held on …………………

Internal Examiner   External Examiner

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1		Design a simple Android app using Kotlin basics. Display a welcome message on the screen. Use variables, data types, and control statements.		
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3		Develop an app to manage student details. Use classes and objects in Kotlin. Display student information on the screen.		
4		Build your first Android app. Add a button and handle click events. Show a toast message when clicked.		
5		Design a login screen using layouts. Use Linear Layout or Constraint Layout. Include username and password fields.		
6		Create navigation between two screens. Move from Login screen to home screen. Use intents or navigation components.		
7		Demonstrate Activity and Fragment lifecycle. Log each lifecycle method. Display lifecycle changes on screen.		
8		Develop an app using MVVM architecture. Create UI layer with View Model. Display a list of items.		
9		Build an app with data persistence. Use Room database to store data. Retrieve and display stored data.		
10		Create an app using Recycler View. Display a list of items with images. Handle item click events.		
11		Develop an app to connect to the internet. Fetch data from an API. Display the fetched data.		
12		Implement Repository Pattern. Use Work manager for background tasks. Fetch and cache data efficiently.		
13		Design a user-friendly app UI. Apply colors, spacing, and alignment. Ensure good usability and user experience.		

Ex.No. 01

WELCOME APP

AIM:

To design a simple Android app using Kotlin basics to display a welcome message using variables, data types, and control statements.

CODE:

MainActivity.kt:

```
package com.example.mad
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 // Make sure this matches your XML file name
 setContentView(R.layout.activity_main)
 val textView = findViewById<TextView>(R.id.textView)
 textView.text = "Welcome Shakthi!"}}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent" android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello"
        android:textSize="26sp"

        app:layout_constraintTop_toTopOf="parent" app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent"/>

    </androidx.constraintlayout.widget.ConstraintLayout>
```

OUTPUT:

RESULT:

The app successfully displays a dynamic welcome message on the screen based on user input or predefined values.

Ex No: 02

ARITHMETIC OPERATIONS APP

AIM:

To develop an Android app to perform basic arithmetic operations using Kotlin functions.

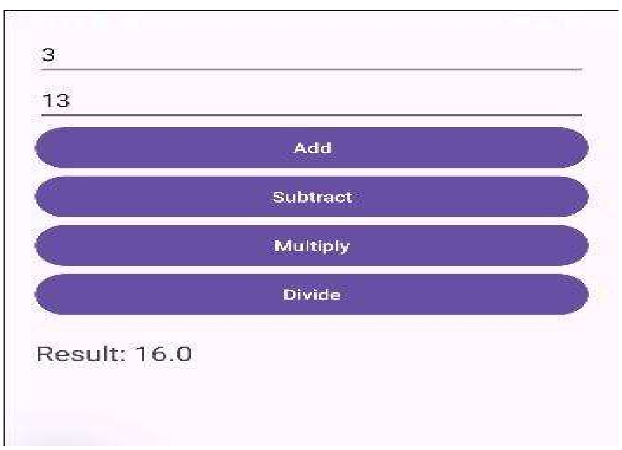
CODE:

MainActivity.kt:

```
package com.example.mad
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
 fun add(a: Double, b: Double) = a + b
 fun sub(a: Double, b: Double) = a - b
 fun mul(a: Double, b: Double) = a * b
 fun div(a: Double, b: Double) = if (b != 0.0) a / b else "Cannot divide by 0"
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 setContentView(R.layout.activity_main)
 val num1 = findViewById<EditText>(R.id.num1)
 val num2 = findViewById<EditText>(R.id.num2)
 val result = findViewById<TextView>(R.id.result)
 val addBtn = findViewById<Button>(R.id.addBtn)
 val subBtn = findViewById<Button>(R.id.subBtn)
 val mulBtn = findViewById<Button>(R.id.mulBtn)
 val divBtn = findViewById<Button>(R.id.divBtn)
 addBtn.setOnClickListener {
 val a = num1.text.toString().toDoubleOrNull()
 val b = num2.text.toString().toDoubleOrNull()
 if (a != null && b != null) {
 result.text = "Result: ${add(a, b)}"
 } else {
 result.text = "Enter valid numbers"
 }
 }
 subBtn.setOnClickListener {
 val a = num1.text.toString().toDoubleOrNull()
 val b = num2.text.toString().toDoubleOrNull()
 if (a != null && b != null) {
 result.text = "Result: ${sub(a, b)}"
 } else {
 result.text = "Enter valid numbers"
 }
 }
```

```
mulBtn.setOnClickListener {
val a = num1.text.toString().toDoubleOrNull()
val b = num2.text.toString().toDoubleOrNull()
if (a != null && b != null) {
result.text = "Result: ${mul(a, b)}"
} else {
result.text = "Enter valid numbers"
}
}
divBtn.setOnClickListener {
val a = num1.text.toString().toDoubleOrNull()
val b = num2.text.toString().toDoubleOrNull()
if (a != null && b != null) {
result.text = "Result: ${div(a, b)}"
} else {
result.text = "Enter valid numbers"
}
}
}
}
```

OUTPUT:



RESULT:

The app successfully performs addition, subtraction, multiplication, and division and displays correct results based on user input.

Ex. No: 03

STUDENT DETAILS

AIM:

To create an Android application using classes and objects to manage and display student details.

CODE:

MainActivity.kt

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 android:layout_width="match_parent"
 android:layout_height="match_parent">
 <TextView
 android:id="@+id/textView"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="Student Details"
 android:textSize="20sp"
 app:layout_constraintTop_toTopOf="parent"
 app:layout_constraintBottom_toBottomOf="parent"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintEnd_toEndOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

OUTPUT:

RESULT:

The app successfully stores student data using a class and displays the details on the screen.

Ex. No: 04

BUTTON CLICK

AIM:

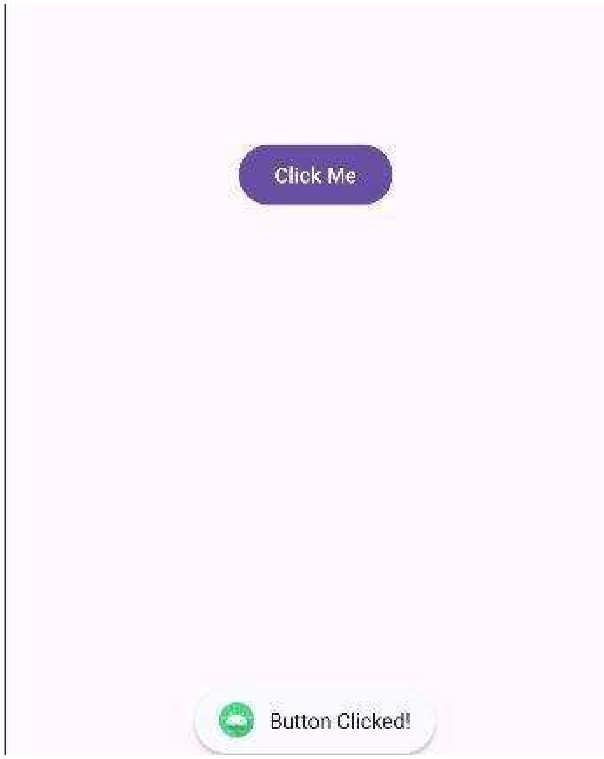
To build an Android app with a button and handle click events to display a Toast message.

CODE:

MainActivity.kt

```
package com.example.mad

import android.os.Bundle

import android.widget.Button

import android.widget.Toast

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 val button = findViewById<Button>(R.id.button)

 button.setOnClickListener {

 Toast.makeText(this, "Button Clicked!", Toast.LENGTH_SHORT).show()

 }

 }

}
```

OUTPUT:



RESULT:

The app successfully shows a Toast message when the button is clicked.

Ex. No: 05

LOGIN SCREEN UI

AIM:

To design a login screen using layouts like LinearLayout or ConstraintLayout with username and password fields.

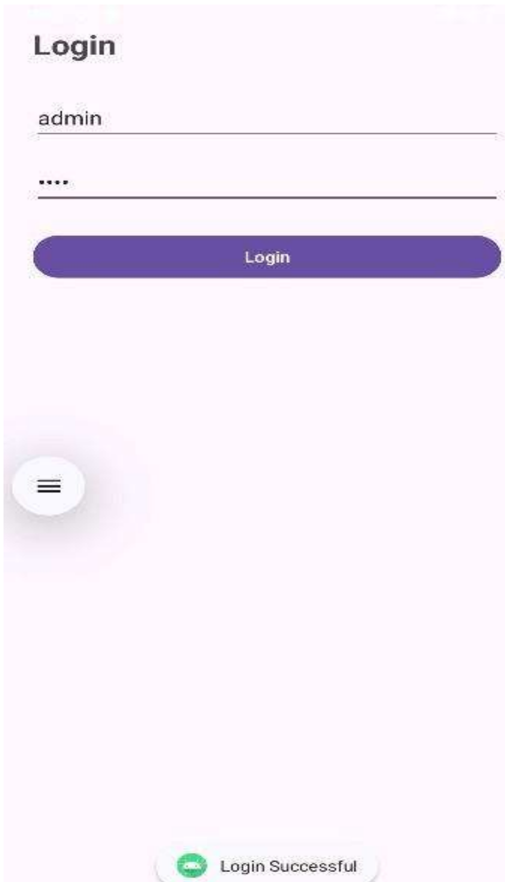
CODE:

MainActivity.kt:

```
package com.example.mad

import android.os.Bundle

import android.widget.*

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 val username = findViewById<EditText>(R.id.username)

 val password = findViewById<EditText>(R.id.password)

 val loginBtn = findViewById<Button>(R.id.loginBtn)

 loginBtn.setOnClickListener {

 val user = username.text.toString()

 val pass = password.text.toString()

 if (user == "admin" && pass == "1234") {

 Toast.makeText(this, "Login Successful", Toast.LENGTH_SHORT).show()

 } else {

 Toast.makeText(this, "Invalid Credentials",

Toast.LENGTH_SHORT).show()

 }

 }
```

```
  }
}
```

OUTPUT:



RESULT:

A user-friendly login interface is created with proper alignment, spacing, and input fields.

Ex. No: 06

NAVIGATION

AIM:

To implement navigation between two screens using intents.

CODE:

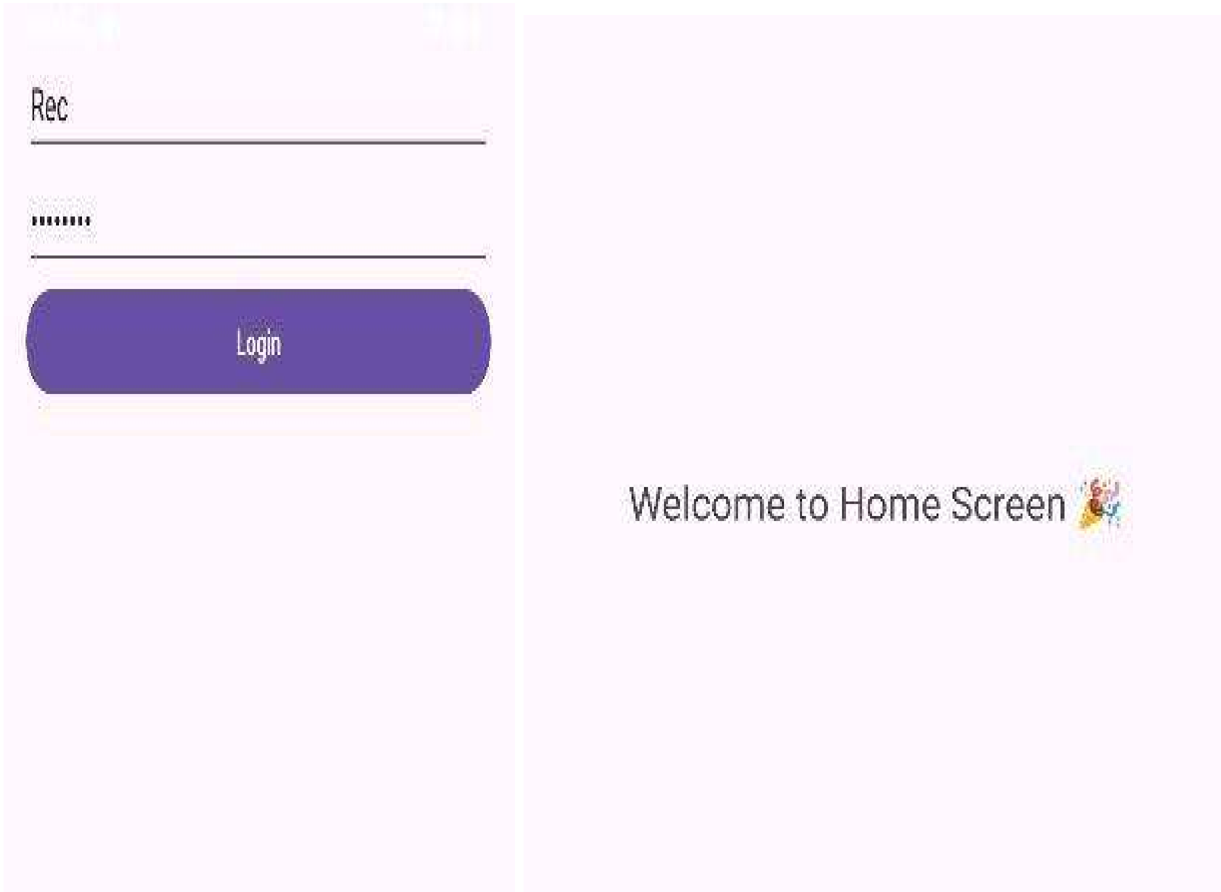
MainActivity.kt:

```
package com.example.mad
import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 setContentView(R.layout.activity_main)
 val loginBtn = findViewById<Button>(R.id.loginBtn)
 loginBtn.setOnClickListener {
 val intent = Intent(this, HomeActivity::class.java)
 startActivity(intent)}}}
```

HomeActivity.kt:

```
package com.example.mad
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
class HomeActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_home)}}
```

OUTPUT:



RESULT:

The app successfully navigates from the login screen to the home screen on button click.

Ex. No: 07

ACTIVITY LIFECYCLE

AIM:

To demonstrate the lifecycle methods of an Activity and observe state changes.

CODE:

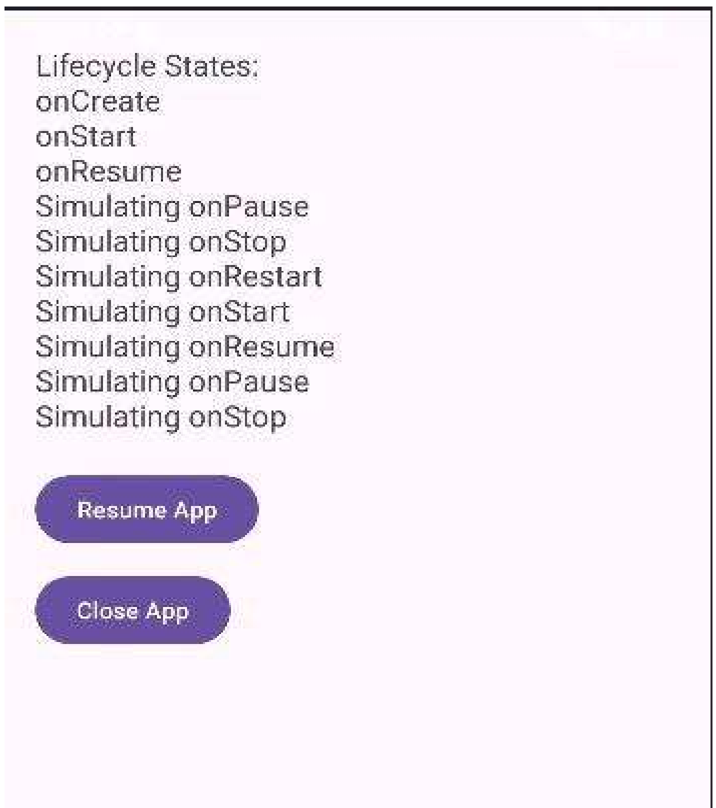
MainActivity.kt

```
package com.example.mad

import android.os.Bundle

import android.util.Log

import android.widget.Button

import android.widget.TextView

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

 lateinit var textView: TextView

 var isPaused = false

 fun updateState(state: String) {

 textView.append("\n$state")

 Log.d("Lifecycle", state)

 }

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 textView = findViewById(R.id.textView)

 val triggerBtn = findViewById<Button>(R.id.triggerBtn)

 val closeBtn = findViewById<Button>(R.id.closeBtn)

 updateState("onCreate")

 // Trigger lifecycle simulation
```

```
triggerBtn.setOnClickListener {
if (!isPaused) {
updateState("Simulating onPause")
updateState("Simulating onStop")
isPaused = true
triggerBtn.text = "Resume App"
} else {
updateState("Simulating onRestart")
updateState("Simulating onStart")
updateState("Simulating onResume")
isPaused = false
triggerBtn.text = "Pause App"}}
// Real close (actual lifecycle)
closeBtn.setOnClickListener {
updateState("Closing App → finish()")
finish()}}
override fun onStart() {
super.onStart()
updateState("onStart")}
override fun onResume() {
super.onResume()
updateState("onResume")}
override fun onPause() {
super.onPause()
updateState("onPause")}
override fun onStop() {
super.onStop()
```

```
updateState("onStop")}

override fun onDestroy() {

super.onDestroy()

Log.d("Lifecycle", "onDestroy")}}
```

OUTPUT:



RESULT:

Lifecycle methods like onCreate(), onStart(), onResume(), onPause(), onStop(), and onDestroy() are successfully logged and observed.

Ex. No: 08

MVVM ARCHITECTURE

AIM:

To develop an Android app using MVVM architecture with ViewModel and LiveData.

CODE:

MainActivity.kt:

```
package com.example.mad
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.lifecycle.ViewModelProvider
class MainActivity : AppCompatActivity() {
 lateinit var viewModel: MyViewModel
 lateinit var textView: TextView
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 setContentView(R.layout.activity_main)
 textView = findViewById(R.id.textView)
 // Get ViewModel
 viewModel = ViewModelProvider(this)[MyViewModel::class.java]
 // Observe LiveData
viewModel.data.observe(this) { list ->
textView.text = list.joinToString("\n")}}}
```

MyViewModel.kt

```
package com.example.mad
import androidx.lifecycle.MutableLiveData
import androidx.lifecycle.ViewModel
```

```
class MyViewModel : ViewModel() {
val data = MutableLiveData<List<String>>()
init {
// Sample data (Model)
data.value = listOf(
"Apple",
"Banana",
"Orange",
"Mango"
)}}
```

OUTPUT:

RESULT:

The app successfully separates UI and data logic and displays a list of items using ViewModel..

Ex. No: 09

ROOM DATABASE

AIM:

To implement local data persistence using Room Database.

CODE:

MainActivity.kt

```
package com.example.mad
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import androidx.room.Room
class MainActivity : AppCompatActivity() {
 lateinit var db: AppDatabase
 lateinit var textView: TextView
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 setContentView(R.layout.activity_main)
 val editText = findViewById<EditText>(R.id.editText)
 val saveBtn = findViewById<Button>(R.id.saveBtn)
 textView = findViewById(R.id.textView)
 db = Room.databaseBuilder(
 applicationContext,
 AppDatabase::class.java, "user-db"
 ).allowMainThreadQueries().build()
 saveBtn.setOnClickListener {
 val name = editText.text.toString()
 if (name.isNotEmpty()) {
```

```
 db.userDao().insert(User(name = name))
 displayData()
 editText.text.clear()}}
 displayData()}}
 fun displayData() {
 val users = db.userDao().getAll()
 textView.text = users.joinToString("\n") { it.name }}}
```

User.kt

```
package com.example.mad
import androidx.room.Entity
import androidx.room.PrimaryKey
@Entity
data class User(
 @PrimaryKey(autoGenerate = true) val id: Int = 0,
 val name: String
)
```

UserDAO.kt

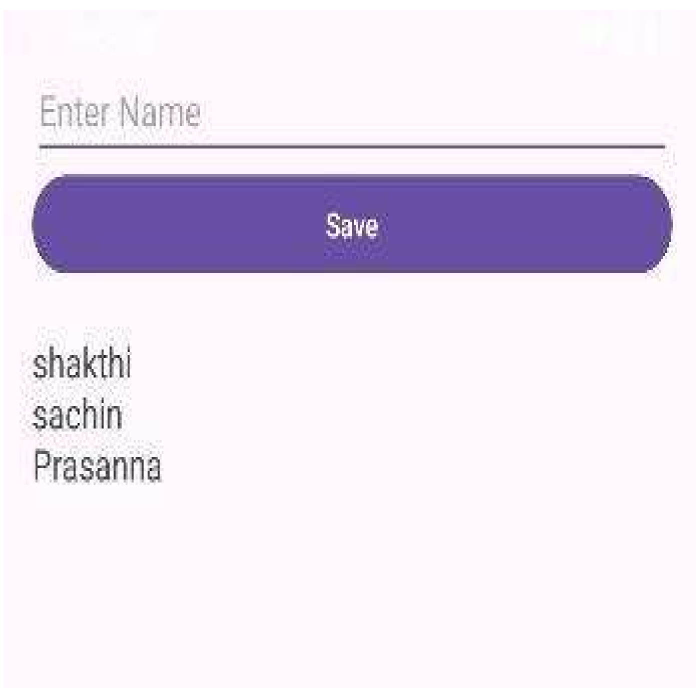
```
package com.example.mad
import androidx.room.Dao
import androidx.room.Insert
import androidx.room.Query
@Dao
interface UserDao {
 @Insert
 fun insert(user: User)
 @Query("SELECT * FROM User")
 fun getAll(): List<User>
```

```
}
```

Appdatabase.kt

```
package com.example.mad

import androidx.room.Database

import androidx.room.RoomDatabase

@Database(entities = [User::class], version = 1)

abstract class AppDatabase : RoomDatabase() {

abstract fun userDao(): UserDao

}
```

OUTPUT:



RESULT:

The app successfully stores, retrieves, and displays data using Room (Entity, DAO, Database).

Ex. No: 10

RECYCLE VIEW

AIM:

To create an app that displays a list of items using RecyclerView and handles item click events..

CODE:

MainActivity.kt

```
package com.example.mad

import android.os.Bundle

import androidx.appcompat.app.AppCompatActivity

import androidx.recyclerview.widget.LinearLayoutManager

import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 val recyclerView = findViewById<RecyclerView>(R.id.recyclerView)

 val list = listOf(

 Item("Apple", R.drawable.apple),

 Item("Banana", R.drawable.apple),

 Item("Orange", R.drawable.apple),

 Item("Mango", R.drawable.apple),

 Item("Grapes", R.drawable.apple)

 )

 recyclerView.layoutManager = LinearLayoutManager(this)

 recyclerView.adapter = MyAdapter(list)

 }

 }
```

MyAdapter.kt

```
package com.example.mad

import android.view.LayoutInflater

import android.view.View

import android.view.ViewGroup

import android.widget.*

import androidx.recyclerview.widget.RecyclerView

class MyAdapter(private val list: List<Item>) :

RecyclerView.Adapter<MyAdapter.ViewHolder>() {

 class ViewHolder(itemView: View) : RecyclerView.ViewHolder(itemView) {

 val textView: TextView = itemView.findViewById(R.id.itemText)

 val imageView: ImageView = itemView.findViewById(R.id.itemImage)

 }

 override fun onCreateViewHolder(parent: ViewGroup, viewType: Int):

ViewHolder {

 val view = LayoutInflater.from(parent.context)

 .inflate(R.layout.item_layout, parent, false)

 return ViewHolder(view)

 }

 override fun onBindViewHolder(holder: ViewHolder, position: Int) {

 val item = list[position]

 holder.textView.text = item.name

 holder.imageView.setImageResource(item.image)

 holder.itemView.setOnClickListener {

 Toast.makeText(

 holder.itemView.context,

 "Clicked: ${item.name}",
```

```
Toast.LENGTH_SHORT
).show()
}
}
override fun getItemCount(): Int = list.size
}
```

OUTPUT:

RESULT:

The app successfully displays a scrollable list with images/text and responds to item clicks.

Ex. No: 11

API FETCH

AIM:

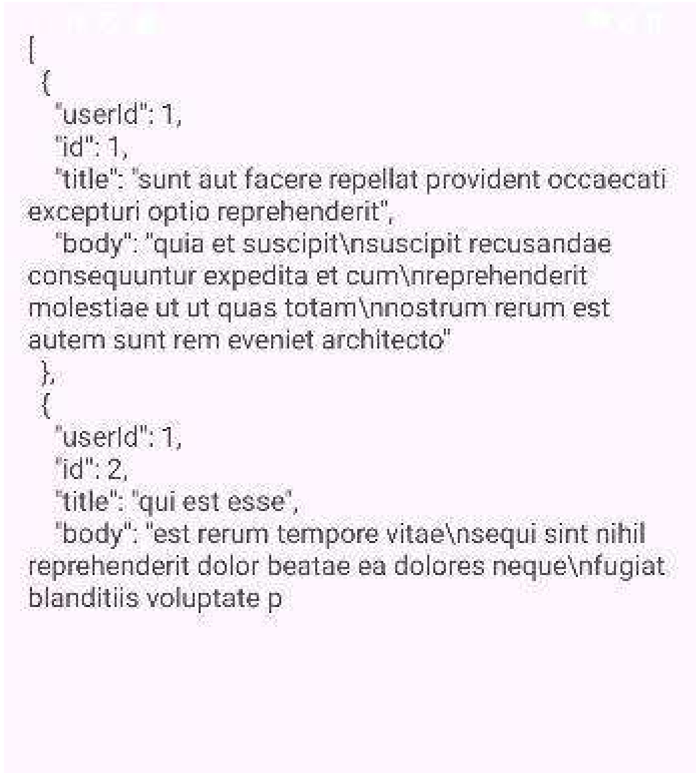
To develop an app that connects to the internet and fetches data from an API.

CODE:

MainActivty.kt:

```
package com.example.mad

import android.os.Bundle

import android.widget.TextView

import androidx.appcompat.app.AppCompatActivity

import java.net.URL

import kotlin.concurrent.thread

class MainActivity : AppCompatActivity() {

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 val textView = findViewById<TextView>(R.id.textView)

 thread {

 try {

 val url = URL("https://jsonplaceholder.typicode.com/posts")

 val result = url.readText()

 runOnUiThread {

 textView.text = result.take(500) // show partial data

 }

 } catch (e: Exception) {

 runOnUiThread {

 textView.text = "Error: ${e.message}"}}}
```

OUTPUT:



RESULT:

The app successfully retrieves data from an API and displays it on the screen.

Ex. No: 12

WORK MANAGER

AIM:

To implement background task execution using WorkManager.

CODE:

MainActivity.kt:

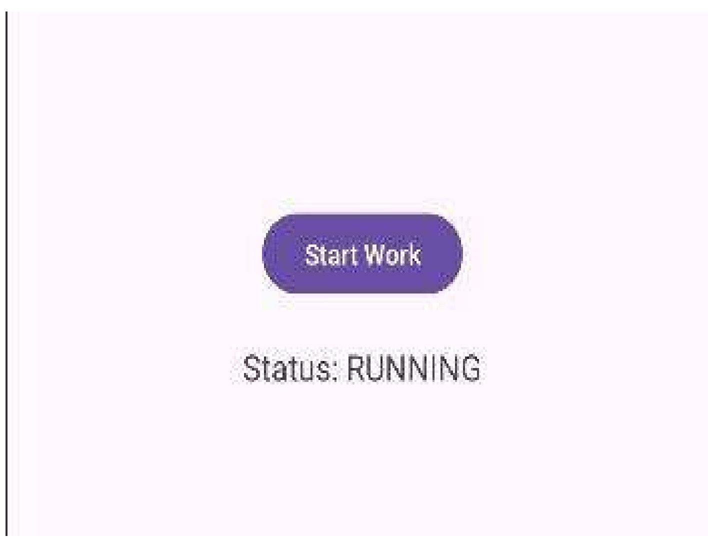
```
package com.example.mad

import android.os.Bundle

import android.widget.Button

import android.widget.TextView

import androidx.appcompat.app.AppCompatActivity

import androidx.work.*

class MainActivity : AppCompatActivity() {

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 val btn = findViewById<Button>(R.id.button)

 val statusText = findViewById<TextView>(R.id.statusText)

 btn.setOnClickListener {

 val workRequest =

OneTimeWorkRequest.Builder(MyWorker::class.java).build()

WorkManager.getInstance(this).enqueue(workRequest)

 // OBSERVE STATUS

 WorkManager.getInstance(this)

 .getWorkInfoByIdLiveData(workRequest.id)

 .observe(this) { workInfo ->

 if (workInfo != null) {
```

```
statusText.text = "Status: ${workInfo.state}"}}}}}
```

MyWorker.kt:

```
package com.example.mad

import android.content.Context

import android.util.Log

import android.widget.Toast

import androidx.work.Worker

import androidx.work.WorkerParameters

class MyWorker(context: Context, params: WorkerParameters)

 : Worker(context, params) {

 override fun doWork(): Result {

 Log.d("WorkManager", "Task Running...")

 Thread.sleep(2000) // simulate work

 Log.d("WorkManager", "Task Completed")

 return Result.success()}}
```

OUTPUT:



RESULT:

The app successfully executes background tasks and updates the UI/log with task status.

Ex. No: 13

UI DESIGN

AIM:

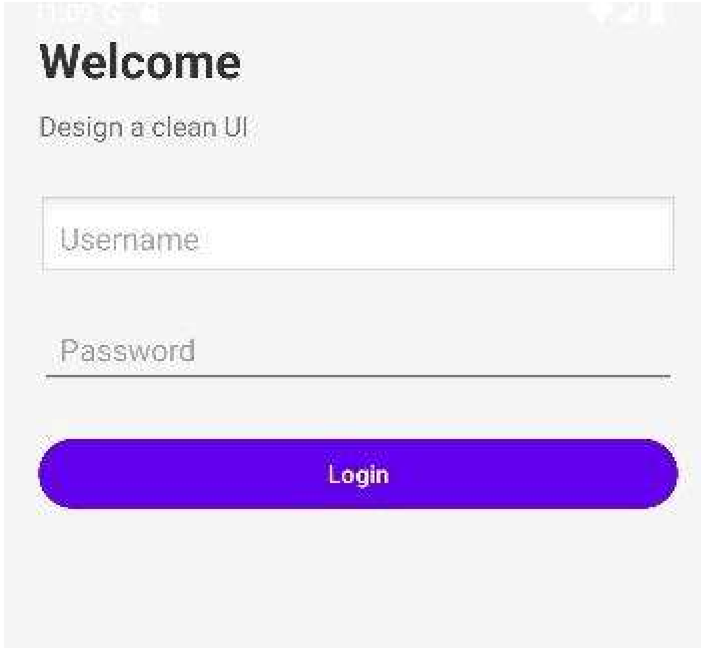
To design a user-friendly Android interface with proper colors, spacing, and alignment.

CODE:

MainActivity.kt:

```
package com.example.mad

import android.os.Bundle

import android.widget.*

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

 override fun onCreate(savedInstanceState: Bundle?) {

 super.onCreate(savedInstanceState)

 setContentView(R.layout.activity_main)

 val btn = findViewById<Button>(R.id.loginBtn)

 btn.setOnClickListener {

 Toast.makeText(this, "UI Designed Successfully!",

Toast.LENGTH_SHORT).show()

 }

 }

}
```

OUTPUT:



RESULT:

A visually appealing and user-friendly UI is created following design principles and usability guidelines.